

SCIENTIFIC COMMUNICATIONS

THE SEARCH FOR THE “MISSING HALF” OF A WORLD-CLASS COPPER RESOURCE IN THE CHUQUICAMATA DISTRICT, CHILE

Adrianna L. Virmond,¹ Andrew J. Tomlinson,² David Selby,³ Dawid Szymanowski,¹ Rodrigo Rojas-Arancibia,⁴ Nicolas Blanco,² and Cyril Chelle-Michou^{1,†}

¹*Institute of Geochemistry and Petrology, ETH Zurich, Zurich 8092, Switzerland*

²*Servicio Nacional de Geología y Minería, Providencia, Santiago 7520405, Chile*

³*Department of Earth Sciences, Durham University, Durham DH1 3LE, United Kingdom*

⁴*Gerencia de Recursos Mineros y Desarrollo, División Radomiro Tomic, CODELCO, Calama 1390000, Chile*

Abstract

Orebodies may be dismembered by post-ore faulting, and partial burial by more recent cover can complicate the acquisition of structural data. This poses a significant challenge to tectonic reconstructions and the discovery of displaced mineralized blocks. A prime example is the behemothian Chuquicamata porphyry copper district in northern Chile. This district exhibits major postmineralization strike-slip faulting that translated a potentially sizable portion of the original deposit to a yet unknown location, likely hidden by thick, postmineralization, post-tectonic sedimentary cover. Since the 1990s, the Ministro Hales deposit, located ~10 km south of Chuquicamata, has been proposed by some investigators as being that piece. Here, we use new high-precision zircon and molybdenite geochronology results to show that although both deposits are derived from the same multimillion-year Chuquicamata Intrusive Complex, they formed 1.5 to 2 Ma apart, ruling out their twinship. However, a consistent southward-younging trend in magmatic and hydrothermal activities across the Chuquicamata Intrusive Complex suggests correlation of the Ministro Hales deposit with the oldest and northernmost part of the Chuquicamata Intrusive Complex, ~10 km north of the Chuquicamata mine. A compilation of available exploration data from beneath the Oligocene to Quaternary sedimentary cover in the Calama basin enabled the construction of a basement geologic map, which independently supports the fault offset inferred from the geochronological correlations. The geologic and geochronological data suggest the “missing half” of Chuquicamata may be located ~20 km south of the Chuquicamata mine, below the City of Calama.

Introduction

Translithospheric strike-slip faults are important controls on arc magma emplacement and related magmatic-hydrothermal ore deposits (Richards, 2003). Critically, postmineralization movements along faults can cause significant displacement and even dismemberment of orebodies, potentially splitting deposits into several disconnected pieces (Stavast et al., 2008; Crane and Kavalieris, 2012; Nickerson and Seedorf, 2016; Sepp and Dilles, 2018). Reconstructing the initial geometry of such strike-slip faults requires extensive geologic field mapping and structural analysis (e.g., Fu and Awata, 2007; Darin and Dorsey, 2013), often aided by geochronological constraints (e.g., Phillips et al., 2004; Wang et al., 2012). However, the absence of good field exposures or displacement markers (e.g., offset guide beds) can significantly complicate the task, especially for systems partially or wholly located below postmineralization cover, which are likely the most promising search areas for new major mineral discoveries (Arndt et al., 2017).

Located in northern Chile, the Eocene-Oligocene Chuquicamata Intrusive Complex is associated with one of the largest porphyry copper systems in the world, hosting >136 million tonnes (Mt) of contained Cu in two main deposits: Chuquicamata and Radomiro Tomic. There, a N-striking strike-slip

fault (the West fault) truncates the Chuquicamata deposit, displacing the western portion of this behemothian deposit to an unknown location (Baker and Guilbert, 1987; Ossandón et al., 2001; Fig. 1). Correlations across the fault point to a net sinistral 35-km horizontal and ~500-m down-on-the-west vertical displacement (Tomlinson et al., 2018), putting the detached piece of Chuquicamata below the thick (up to >400 m) Oligocene to Holocene sediments of the Calama basin. Yet exploration campaigns 35 km south of the Chuquicamata mine have proven fruitless (Zentilli et al., 2021).

Since the 1990s, several investigators have discussed whether the Ministro Hales deposit, located ~10 km south of Chuquicamata on the western side of the fault system, could be its “missing half” (Alvarez, 1993; Sillitoe et al., 1996; McInnes et al., 1999; Tomlinson et al., 2001; Zentilli et al., 2015, 2018, 2021). Bulk-rock composition, radiogenic isotopes, fluid inclusions, basement rocks, and their locations on either side of the West fault all suggest a genetic relationship between the Ministro Hales and Chuquicamata deposits (e.g., Alvarez, 1993; Zentilli et al., 2015, 2018, 2021). However, apparent age differences (mostly from a scarce data set in internal company reports cited in Rivera et al., 2012), partially distinct alteration and mineralization styles (e.g., Sillitoe et al., 1996), and the relatively short distance between the two deposits (~10 km) raise uncertainties concerning the exact nature of their link.

[†]Corresponding author: email, cyril.chelle-michou@eaps.ethz.ch

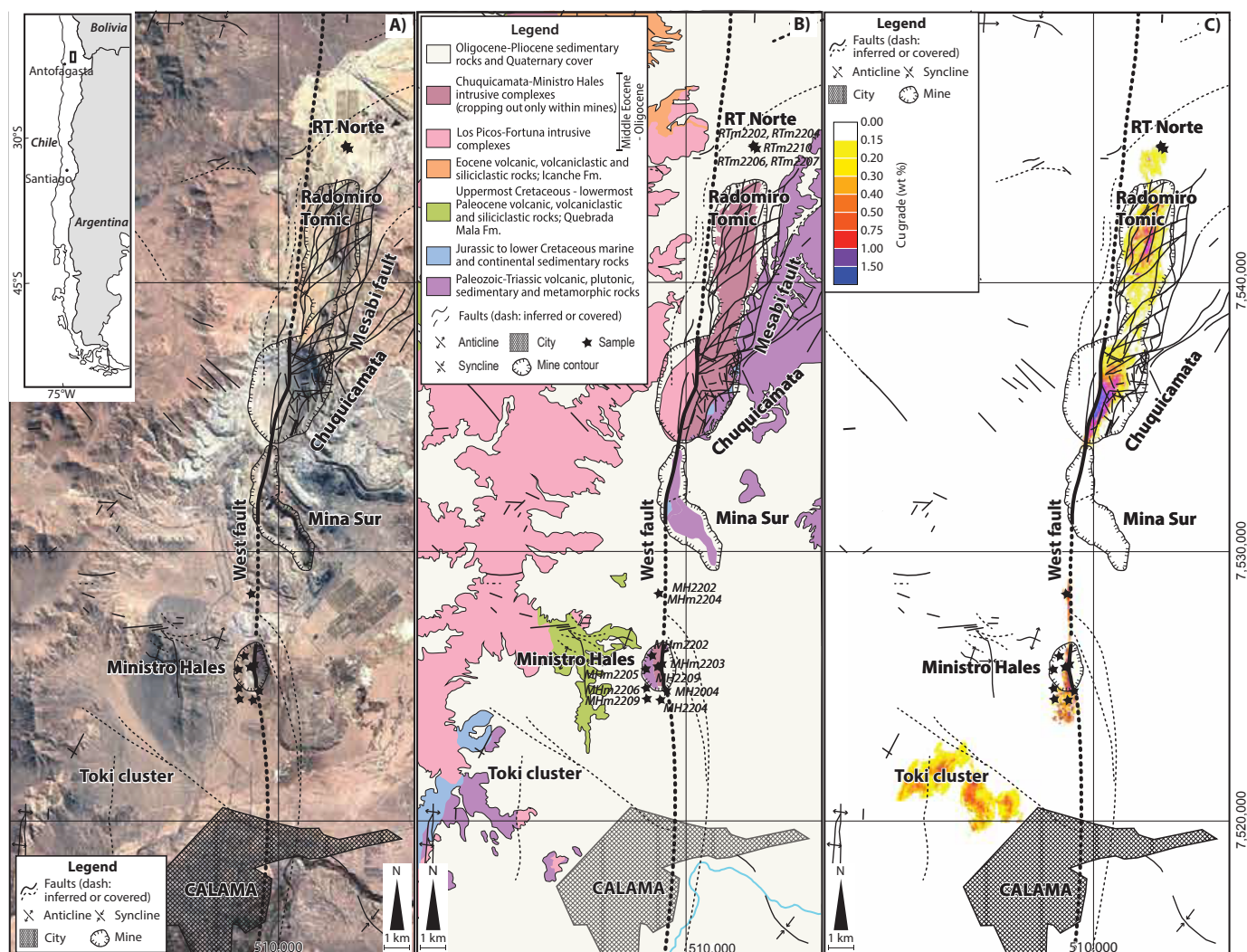


Fig. 1. Location of the Ministro Hales deposit and the Chuquicamata Intrusive Complex (CIC). (A) Satellite image with annotated structures. (B) Simplified geologic map of the district, after Rivera et al. (2012) and Tomlinson et al. (2018). Note that part of the Ministro Hales Intrusive Complex exposed at the surface is not visible at the map scale, and all samples are from the subsurface. (C) Copper grade model map of the district. Assembled from level 2,250 m above sea level (m a.s.l.) for RT Norte, Radomiro Tomic, and Chuquicamata; level 2,000 m a.s.l. for Ministro Hales; and level 1,800 m a.s.l. for the Toki cluster.

Here, we use new and published (Virmond et al., 2024) zircon and molybdenite high-precision geochronological data to test the temporal relationship between the Chuquicamata and Ministro Hales deposits. We show that the magmatic and hydrothermal histories of both deposits were not synchronous, thereby excluding a direct spatial and temporal correlation between the two deposits. Instead, we show that Ministro Hales has greater temporal affinity and geologic similarities with the RT Norte deposit, located 10 km north of the Chuquicamata mine. This has key implications for interpreting the amount of displacement along the West fault, and the possible location of the “missing half” of Chuquicamata. The development of a new geologic map of the basement beneath the sedimentary deposits of the Calama basin, combined with a compilation of available exploration data, offers additional insights into the potential location of this elusive orebody.

Geologic Context

The Eocene-Oligocene contractional period of the Andean tectonic cycle was characterized by the eastward migration of the magmatic arc and the development of a north-south metallogenic belt (Camus and Dilles, 2001; Cobbold et al., 2007). The N-S-trending Domeyko fault system is a major transcurrent structural system located in northern Chile, active since the middle Eocene, during Andean compression, and is considered a first-order control on the emplacement of major porphyry systems in the area, including the Chuquicamata and Ministro Hales deposits (Maksaev and Zentilli, 1999; Camus and Dilles, 2001; Amilibia et al., 2008; Mpodozis and Cornejo, 2012).

The Chuquicamata district (Fig. 1) is part of the Eocene-Oligocene metallogenic belt of the Central Andes and comprises several porphyry copper deposits (Sillitoe, 1988; Sillitoe and Perello, 2005). The West fault, the local name of the

Domeyko system, splits the district into two blocks: the eastern, where the Chuquicamata (with its associated Mina Sur exotic copper deposit), Radomiro Tomic, and RT Norte Cu-Mo deposits are located (all genetically related to the Chuquicamata Intrusive Complex), and the western block, hosting the Ministro Hales deposit, the Eocene Fortuna Granodiorite Complex and associated Toki cluster of Cu-Mo deposits (Toki, Quetena, Genoveva, Miranda, Opache, and Pulucktur), and the precursor Eocene Los Picos Igneous Complex (Ossandón et al., 2001; Rivera et al., 2012). Activity of the West fault was protracted and polyphasic. A proto-West fault system coincided with the emplacement of the Chuquicamata Intrusive Complex in a dextral transcurrent setting defined by the Mesabi fault, exposed on the east side of the Chuquicamata Intrusive Complex (Fig. 1), and a hypothetical proto-West fault situated either in the approximate position of the current West fault or in some position west of it (Lindsay et al., 1995; Tomlinson et al., 2018; Rojas-Arancibia, 2023). During the Oligocene to early Miocene, the kinematics of the system became sinistral, resulting in over 35 km of displacement of geologic units across it (Dilles et al., 1997; Tomlinson and Blanco, 1997; Rivera et al., 2012). A final phase of activity in the middle Miocene-Pliocene resulted in <1 km of dextral offset (Tomlinson et al., 2018).

At Chuquicamata, the typical porphyry-style alteration pattern is only developed to the east of the West fault and terminates sharply against it, lacking a symmetrical alteration zonation typical of other deposits in the district and globally (e.g., Sillitoe, 1973; Ossandón et al., 2001). Furthermore, the distribution of copper grades at the deposit scale indicates that the Chuquicamata orebody, which is slightly oblique to the West fault, is sharply interrupted against the fault, beyond which the Fortuna granodiorite is barren (e.g., Sillitoe, 1973; Ossandón et al., 2001; Fig. 1C). These features, together with observations by mine geologists of mineralized rock fragments (both hypogene and supergene) in the fault gouge, are the basis supporting that a potentially sizable portion of the original deposit has been offset by the West fault.

Geology of the Ministro Hales deposit

The Ministro Hales Intrusive Complex is a granodioritic complex, composed of dikes and stocks associated with the mineralization and hydrothermal alteration of the Ministro Hales deposit (Sillitoe et al., 1996; Rivera et al., 2012). The deposit occurs alongside and parallel to the West fault as a north-south orebody extending up to 2 km south and 4 km north of the current open pit. The complex is covered by Miocene-Holocene alluvial deposits, intrudes the Triassic Mansa Mina granodiorite to the west, and is truncated by the West fault on its east side, where it is in contact with the lower Oligocene-lower Miocene Calama Formation conglomerates (Sillitoe et al., 1996; Boric et al., 2009; Rivera et al., 2012; Tomlinson et al., 2018). Intrusive phases include the volumetrically dominant equigranular to porphyritic Mansa Mina porphyry, which is intruded at depth by quartz porphyry and quartz-poor dacitic porphyry dikes (also mapped as feldspar porphyry; e.g., Boric et al., 2009; Zentilli et al., 2015, 2018).

The Ministro Hales deposit contains 25.9 Mt of Cu (past production + resources from Camus and Castelli, 2021, and CODELCO, 2022, respectively). It is described as a classic

porphyry system with Cu-Mo-mineralized potassic alteration, telescoped by high-sulfidation veins in the shallow parts of the deposit with phyllic (sericite) and advanced argillic (with alunite, pyrophyllite, sericite, and dickite) alteration, and hydrothermal breccias rooted in the Mansa Mina porphyry (Sillitoe et al., 1996; Boric et al., 2009; Rivera et al., 2012; Zentilli et al., 2021). The potassic alteration zone is associated with bornite-chalcocopyrite $\pm$ digenite and molybdenite mineralization (Boric et al., 2009). The hydrothermal breccias have high contents of Cu, Ag, As, and Zn with assemblages that vary from bornite-chalcocite-pyrite at depth, to chalcocopyrite-tennantite-pyrite, and chalcocite-enargite-pyrite $\pm$ bornite $\pm$ covellite closer to the surface (Boric et al., 2009; Zentilli et al., 2021). A thin (a few tens of meters) copper oxide zone (with chrysocolla, chenevixite, and minor malachite and atacamite) with remaining sulfides occurs below a ~50-m-thick leached zone (Sillitoe et al., 1996), which is underlain by an immature secondary sulfide enrichment zone where chalcocite coats hypogene sulfides.

Geology of the RT Norte deposit

The RT Norte deposit lies >500 m north of the Radomiro Tomic open pit, under an up to 200-m-thick Miocene cover (R. Rojas-Arancibia and A. Maldonado, unpub. report, 2020). The RT Norte deposit is located ~1,200 m east of the West fault, situated east of what is known as the barren corridor. The RT Norte system is characterized by mineralized structures 10 to 20 m thick, with a general attitude N20°W/65°SW, 500-m length in plan view, and 300- to 450-m depth (Rojas-Arancibia, 2023). These structures are hosted within porphyries of the Chuquicamata Intrusive Complex, which intrude the Triassic Elena granodiorite. The porphyries of the complex have a granodioritic composition and a common mineralogy and are classified according to their textural characteristics and age (see Virmond et al., 2024). The Chuquicamata Intrusive Complex intrusions are broadly similar (in mineralogy and geochemistry) to porphyries of the Ministro Hales Intrusive Complex (Zentilli et al., 2018), except for the dacite porphyry, which appears to have no equivalent in the Chuquicamata Intrusive Complex.

In the hypogene zone, the Chuquicamata Intrusive Complex at RT Norte presents a classic background potassic alteration associated with quartz-chalcocopyrite-bornite-molybdenite-bearing veins. It is overprinted by structurally controlled phyllic (sericite) and less common advanced argillic (alunite, pyrophyllite, sericite) alteration associated with veins and minor breccia carrying chalcocopyrite overprinted by pyrite-enargite with minor hypogene tennantite-tetrahedrite-covellite-chalcocite and Ag and Au geochemical anomalies (R. Rojas-Arancibia and A. Maldonado, unpub. report, 2020). The supergene oxidized zone is characterized by the presence of brochantite, atacamite, chrysocolla, and copper-bearing clays. A thin supergene-enriched sulfide horizon is also recognized, which is a few tens of meters thick in the vein zone (Rojas-Arancibia, 2023).

Chuquicamata Intrusive Complex High-Precision Geochronological Data

An extensive high-precision geochronological data set has been presented for the Chuquicamata Intrusive Complex and associated mineralization at Chuquicamata, Radomiro Tomic,

and RT Norte (Virmond et al., 2024). This data set includes chemical abrasion-isotope dilution-thermal ionization mass spectrometry (CA-ID-TIMS) U-Pb dating of 234 individual zircon grains representing 18 porphyry samples together with 18 new molybdenite Re-Os dates complementing nine previously reported molybdenite Re-Os dates (Mathur et al., 2000; Barra et al., 2013; Zentilli et al., 2021).

These data reveal that the Chuquicamata Intrusive Complex was assembled in two phases. The old porphyries of the Chuquicamata Intrusive Complex were emplaced between 35.221 ± 0.059 and 34.402 ± 0.053 Ma (all dates are reported at the 2σ uncertainty level including analytical, tracer calibration, and decay constant uncertainty), with dates around 35 Ma being characteristic of the Radomiro Tomic and RT Norte sectors, while dates around 34.5 Ma were determined at Chuquicamata (see Fig. 2B). After a ~ 0.5 Ma-long magmatic lull, the young porphyries intruded first in RT Norte at 33.880 ± 0.054 Ma, 33.723 ± 0.052 Ma at Radomiro Tomic, 33.624 ± 0.080 Ma in the bridge between the Radomiro Tomic and Chuquicamata open pits, and 33.249 ± 0.063 at Chuquicamata (Fig. 2B). It is noteworthy that the historical nomenclature of porphyries of the Chuquicamata Intrusive Complex (i.e., Este, Oeste, Fino, and Banco porphyries), which is based on textural features, did not correlate with their emplacement ages. In several instances, porphyries mapped as the same unit would belong to either the old or the young porphyries as revealed by their respective emplacement ages (Virmond et al., 2024). Molybdenite Re-Os dates also highlight the southward younging of the hydrothermal activity with dates of 34.98 ± 0.18 and 34.19 ± 0.18 Ma at RT Norte, 34.19 ± 0.17 to 33.07 ± 0.17 Ma at Radomiro Tomic, and 32.74 ± 0.19 to 31.52 ± 0.16 Ma at Chuquicamata; in general, the Re-Os molybdenite dates are within uncertainty of, or slightly younger than, the emplacement ages of the young porphyries within each deposit (Fig. 2B).

The Chuquicamata Intrusive Complex was emplaced syn-kinematically in a releasing bend or step-over between the Mesabi fault and the proto-West fault (Lindsay et al., 1995; Tomlinson et al., 2018). In the context of this model, the successive southward younging of emplacement of the old porphyries, of the young porphyries, and of the hydrothermal activity suggests the bend or step-over evolved into a trans-tensional duplex that propagated from north to south with multiple linking faults (Fig. 3). After emplacement of the old porphyries in a particular segment, the subsequent history of young porphyry intrusion and mineralization took a similar, but slightly time-offset course for each segment, presumably related to the internal dynamics of the magmatic-hydrothermal system. Activity along the dextral transverse linking faults continued into the mineralization stage, controlling the distribution of hypogene ore for large parts of the Chuquicamata Intrusive Complex (Rojas-Arancibia, 2023).

Exploration Between Calama and Sierra Limón Verde (Ricardo Project)

The most significant displacement event on the West fault, resulting in a net left-lateral offset on the system, is the postmineralization, Oligocene-early Miocene sinistral event (Lindsay et al., 1995; Dilles et al., 1997; Tomlinson et al., 2018). The idea that there might be a sinistrally offset piece of the

Chuquicamata deposit initially came about when Anaconda geologists in the late 1960s intercepted a 60-m-thick sliver of copper-mineralized rock along the West fault while drilling west of the Mina Sur exotic deposit, south of Chuquicamata (J.C. Marquardt, C. Münchmeyer, and J. Proffett, writ. commun., 2025; Fig. 1). The fault sliver contained enargite-pyrite-chalcocite-covellite mineralization, identical to that in the west part of the Chuquicamata deposit from which it was inferred to have been displaced. Subsequent mapping by Richard Baker led to a hypothesis of 25 to 30 km of sinistral offset on the West fault, placing the missing “half” of Chuquicamata south of Calama, between the El Loa airport and the Sierra Limón Verde (Baker and Guilbert, 1987; see Figs. 1 and 4 for location). The Minera Doña Inés company conducted the first drilling program to test Baker’s concept in the early 1980s under the direction of John Hunt. Although it failed to find the missing half of the Chuquicamata deposit, it discovered the Opaque deposit, immediately west of Calama, demonstrating the potential for deposits concealed beneath the postmineral cover of the Calama basin (Green and Grieg, 2013; C. Münchmeyer, writ. commun., 2025). In the late 1980s, Richard Baker and associates consolidated the mining rights to form the Ricardo project, covering an $\sim 15 \times 15$ -km area centered on the West fault and extending from the northern limits of the City of Calama southward to the foothills of the Sierra Limón Verde (Fig. 4). Exploration of the Ricardo project has pursued both the search for the missing half of Chuquicamata in proximity to the West fault and Opaque or Toki-like targets, farther from the West fault.

Exploration of the property was carried out by seven companies in eight separate drilling campaigns and included geochemical and geophysical surveys (gravity, induced polarization, magnetic, and magnetotelluric; Green and Grieg, 2013). As of 2022, a total of 85 holes have been drilled on the property (including water wells), totaling 32,600 m, of which 59 holes can be interpreted to have penetrated into the basement over a total of 10,095 m (Fig. 4). However, many of these holes penetrated only a short distance into the basement (median of 113 m). Although no ore-grade mineralization was discovered, significant porphyry-style hydrothermal alteration (sericitic, propylitic, argillic, and potassic) with associated sulfides (mostly pyrite), veining, and Cu geochemical anomalies was encountered within the basement of the Ricardo property (Green and Grieg, 2013).

Methods

Representative samples of the porphyry phases within the Ministro Hales deposit, as well as a sample of the Mansa Mina porphyry from the Ministro Hales north body (Sillitoe et al., 1996), about 3.5 km north of the mine, were collected for geochronology (Fig. 1; App. Fig. A1; App. Table A1). Zircon separates were obtained by conventional methods, selected grains were mounted in epoxy, and their internal textures were imaged by cathodoluminescence (CL). Zircon grains were first screened for their U-Pb dates by laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS), and selected grains were subsequently extracted from mounts and dated by CA-ID-TIMS at ETH Zurich. Nine quartz-molybdenite veins from RT Norte ($n = 3$) and Ministro Hales ($n = 6$) were selected for Re-Os geochronology,

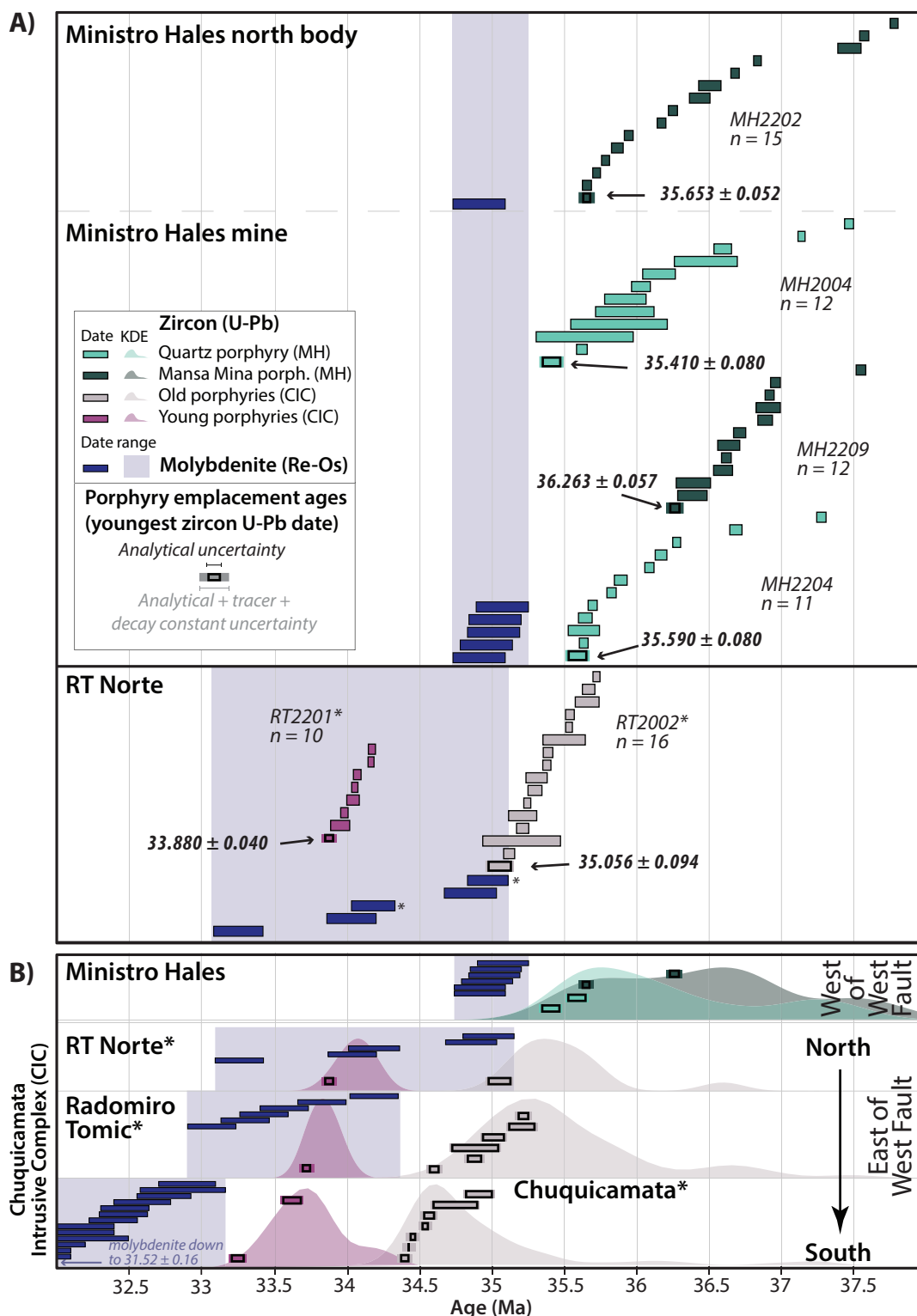


Fig. 2. (A) Geochronological data for the Ministro Hales and RT Norte deposits showing ranked zircon chemical abrasion-isotope dilution-thermal ionization mass spectrometry (CA-ID-TIMS) U-Pb and molybdenite negative-thermal ionization mass spectrometry Re-Os dates. Each bar represents a single-crystal zircon $^{206}\text{Pb}/^{238}\text{U}$ date or a Re-Os molybdenite date and their associated 2σ uncertainty. *Denotes data from Virmond et al. (2024). (B) Compiled geochronological data for each deposit of the Chuquicamata and Ministro Hales Igneous Complexes (Virmond et al., 2024; this study) showing kernel density estimate curves of pooled single-grain zircon CA-ID-TIMS U-Pb data, zircon U-Pb emplacement ages of individual porphyry samples, and molybdenite Re-Os dates. Abbreviations: CA-ID-TIMS = chemical abrasion-isotope dilution-thermal ionization mass spectrometry, MH = Ministro Hales, RT = Radomiro Tomic.

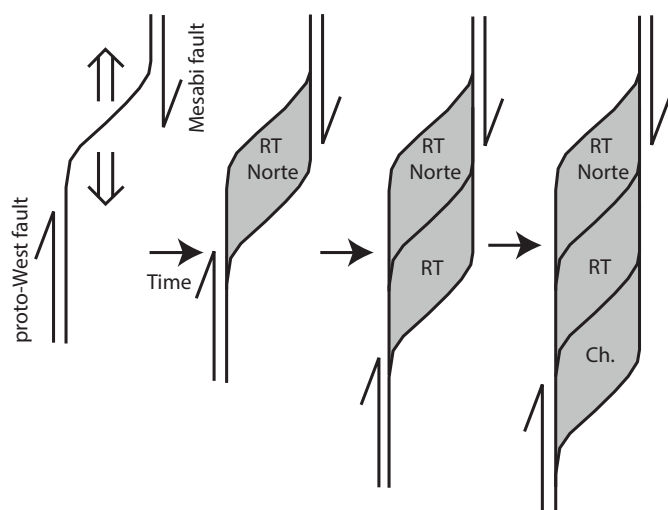


Fig. 3. Model for incremental emplacement of Chuquicamata Intrusive Complex in a dextral transtensional duplex. Abbreviations: Ch. = Chuquicamata, RT = Radomiro Tomic.

and molybdenite was isolated by HF dissolution (Lawley and Selby, 2012) and further purified by handpicking. Molybdenite Re-Os dates were obtained by negative-thermal ionization mass spectrometry (N-TIMS) at Durham University. Individual U-Pb zircon dates are reported with analytical 2σ uncertainty, and the interpreted porphyry emplacement ages additionally include the uncertainty on the calibration of the tracer solution and on the decay constant. Molybdenite Re-Os dates are reported with 2σ uncertainty, including both analytical and decay constant uncertainties. All analytical procedures are detailed in Appendix 1, and results are tabulated in Appendix Tables A2 through A7.

We compiled available subsurface geologic and geophysical data from the Ricardo property and surrounding areas. The majority of the data were obtained from Green and Grieg (2013), augmented and locally corrected using the CODELCO drill hole database, the SIGEX database of the Chilean Geological Survey (SERNAGEOMIN), Pizarro (2011), and personal communications from C. Núñez, C. Münchmeyer, and J.C. Marquardt (2025). We also consulted some of the original drill logs and additionally conducted a firsthand description of the core from 12 drill holes. The available information is limited and sparse, and the absence of a reported feature (e.g., alteration or sulfide minerals) may not preclude its actual presence. In addition to the 85 drill holes from the Ricardo property, an additional 62 holes exist within its immediate vicinity, adding a further ~18,440 m. Geochronological data across the study area are also compiled, including 41 subsurface samples (see App. Tables A8-A11). This material is used to prepare a simplified map of the pre-Oligocene surface and subsurface geology of the Calama basin.

Results

Geochronology

In situ LA-ICP-MS dating revealed that six out of 10 samples mapped as Mansa Mina quartz porphyry or dacitic porphy-

ry were Triassic in age, with weighted mean ages of 235 to 225 Ma. This echoes the mineralogical and textural similarities between the Triassic and Eocene magmatic rocks across the district (Zentilli et al., 2018). Within the four Eocene-age samples, 4 to 7% of the zircon grain cores are inherited or xenocrystic (early Eocene or Triassic).

Four porphyry samples were dated by CA-ID-TIMS (total of 58 grains). Each sample shows a main zircon population spanning 0.7 to 1.2 Ma and collectively record about 1.4 Ma of uninterrupted dates (between ca. 37 and 35.4 Ma; Fig. 2A). Out of these samples, nine older outlier dates (between 39.4 and 37.1 Ma, two of which are >38 Ma and so plot off Fig. 2A) are interpreted to likely include an inherited component not recognized by CL imaging.

The Mansa Mina porphyry samples yield dates between 36.834 ± 0.026 and 35.653 ± 0.032 Ma for a sample collected from the Ministro Hales north body, and between 36.958 ± 0.034 and 36.263 ± 0.040 Ma from the mine itself (Fig. 2A). The two quartz porphyry samples yield similar date distributions, ranging from 36.594 ± 0.061 to 35.410 ± 0.069 Ma.

All the Re-Os molybdenite dates obtained for Ministro Hales overlap within uncertainty and range between 35.08 ± 0.18 and 34.92 ± 0.18 Ma (Fig. 2A). At RT Norte, the new molybdenite dates range between 34.86 ± 0.18 and 33.26 ± 0.17 Ma (Fig. 2A).

Pre-Oligocene geology of the Calama basin

The West fault system in the Ricardo project area is structurally complex, consisting of the master West fault and subsidiary East fault from Ministro Hales to south of the El Loa airport (Fig. 4).

Depth to basement data indicate these two faults bound a north-south graben with over 700 m of Oligocene-Quaternary sedimentary fill in the vicinity of Ministro Hales, over 500 m south of the Loa River, and an intervening area with 100 to 300 m near Calama (Fig. 4). Five holes drilled in the graben south of the Loa River failed to reach basement, leaving the basement unit there undetermined, despite four of the holes having lengths of 504 to 606 m. In the Sierra Limón Verde, the West fault system consists of the Limón Verde and Moctezuma faults. The Limón Verde fault has considerable apparent offset, whereas the Moctezuma fault does not, suggesting the former is the master fault and southern continuation of the West fault.

Although the subsurface bedrock geology is interpretative, some broad features supported by the surface geology stand out. One is the correlation of Eocene Icanche Formation andesitic and dacitic volcanic rocks and breccias cropping out in the Calama and Milagro hills, east of the West fault, with those in the Pampa Limón Verde, west of the West fault (Fig. 4; M. García, unpub. report, 2009; Duhart et al., 2018; Tomlinson et al., 2018). The description of fresh volcanic rocks in drill holes and the available geochronology on core samples (App. Tables A8, A10) suggest the Icanche Formation is the principal subsurface bedrock unit in the areas surrounding these surface exposures. In both areas, the surface exposures coincide with aeromagnetic highs in total magnetic intensity maps (Freeport McMoran South America Ltda., 2019), a relationship that was used, in combination with the core descriptions, to infer the subsurface extent of the unit (Fig. 4).

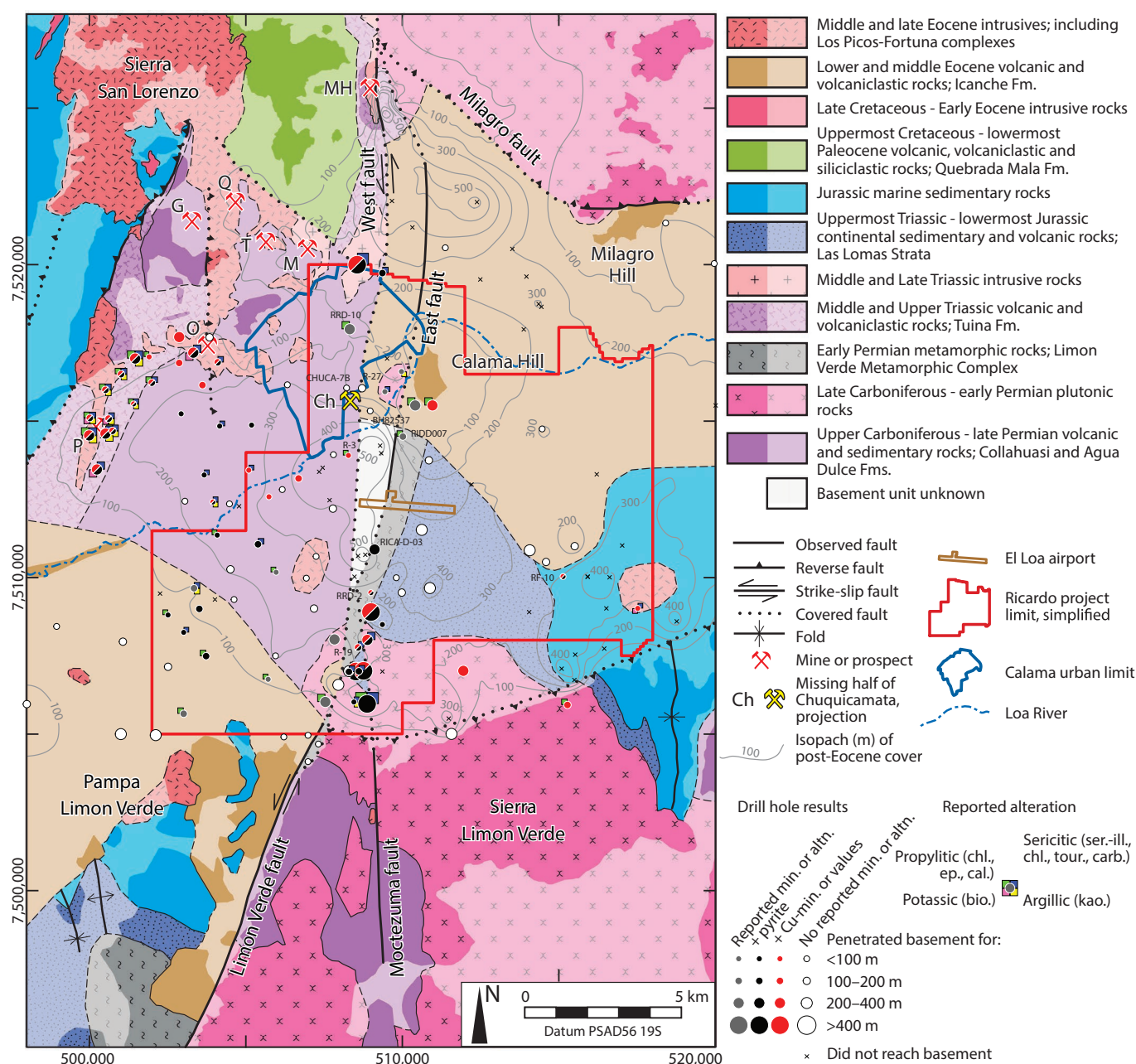


Fig. 4. Simplified map of pre-Oligocene geology for the Ricardo project area, showing location of drill holes, mines, prospects, and projected location of the missing half of the Chuquicamata ore deposit (Ch). Saturated unit colors are surface exposures, and muted colors are subsurface units. Surface geology simplified from Mpodozis et al. (1993), Duhart et al. (2018), and Tomlinson et al. (2018). Subsurface geology in the northwest and north-central part of the map is modified from Rivera et al. (2012). Isopach contours of the post-Eocene cover are interpolated from our compilation of drill hole data (App. Table A8) and the CODELCO drill hole database. Drill hole names are only shown for those mentioned in the text. Available Cu values are considered anomalous (and the corresponding drill hole shown in red) when reaching over 500 ppm for at least 1 m, 300 ppm over 3 m, 100 ppm over 20 m, or over 100 ppm in the case of a spot core sample during rotary drilling (typically 3 m in length). See Appendix Figure A5 for all drill hole names and geochronological data. Drill hole information and geochronology, including data sources, are given in Appendix Tables A8 to A11. Abbreviations: altn. = alteration, bio. = biotite, cal. = calcite, carb. = carbonate, chl. = chlorite, ep. = epidote, G = Genoveva, M = Miranda, MH = Ministro Hales mine, min. = mineralization, O = Opache, P = Pulucktur, Q = Quetena, ser. = sericite, T = Toki, tour. = tourmaline.

Particularly significant is the description of unaltered andesite in holes R-27 and BH82537 located on the southeast margin of the Calama City urban limit (Fig. 4), with an andesite sample from R-27 returning a whole-rock K-Ar date of 46.5

± 2.9 Ma (C. Münchmeyer and J.C. Marquardt, writ. commun., 2025; App. Fig. A5), suggesting the Icanche Formation is contiguous across the East fault. Thickness changes in Neogene strata across the East fault and surface structures along

its trace indicate the East fault was active in the Miocene-Pliocene as a down-to-the-west, right-lateral oblique fault (Jordan et al., 2015; Tomlinson et al., 2018). However, the occurrence of Eocene volcanic rocks on both sides of the East fault implies that the net post-Eocene lateral offset is limited (Fig. 4).

In the Pampa Limón Verde area, the Icanche Formation unconformably overlies a folded uppermost Triassic to Jurassic sedimentary and lesser volcanic section that rests on high-grade Permian metamorphic rocks (Mpodozis et al., 1993; M. Garcia, unpub. report, 2009; Duhart et al., 2018). The Permian metamorphic unit consists of quartz-muscovite-biotite-garnet schist and garnet amphibolite, whereas the Mesozoic section consists of an uppermost Triassic-lowermost Jurassic unit of continental red sandstone, conglomerate, and siltstone, with interbedded silicic tuff and andesite lava, conformably overlain by a Jurassic marine sequence of fossiliferous limestone and calcareous sandstone, siltstone, and shale. A similar Mesozoic section overlies late Paleozoic granite in the Sierra Limón Verde (Mpodozis et al., 1993; M. Garcia, unpub. report, 2009) and is recognized in various drill holes north of Sierra Limón Verde (App. Table A8). Within the lower continental part, both the Pampa Limón Verde and Sierra Limón Verde sections contain 199 to 198 Ma silicic tuffs that are correlative by lithology as well as age (M. Garcia, unpub. report, 2009; App. Table A10). Although high-grade metamorphic rocks are not exposed in the Sierra Limón Verde east of the Limón Verde fault, biotite-muscovite schists were intercepted in four drill holes (RIDD007, RICA-D-03, RRD-2, and R-19) in the Ricardo project and define a NNE-SSW-trending belt located immediately east of the East fault, both north and south of the El Loa airport (Fig. 4). This metamorphic belt coincides with a local NNE-SSW-trending gravity high in a ground-based gravity survey (Green and Greig, 2013; Freeport McMoran South America Ltda., 2019). Biotite schist was also intercepted below Jurassic sandstone in the southeast corner of the Ricardo project (drill hole RF-10; Green and Greig, 2013), suggesting Paleozoic metamorphic rocks may underlie a large part of the Mesozoic section within this part of the Calama basin.

The above-mentioned Eocene and Permian to Jurassic basement domains have approximately 20 km of left-lateral offset across the West and Limón Verde faults (Fig. 4). But restoring this offset also places upper Carboniferous-lower Permian andesitic to rhyolitic volcanic rocks of the Collahuasi Formation, exposed in the vicinity of Calama City and intercepted widely in drill holes west of the West fault, against coeval dioritic to granitic plutonic and metaplutonic rocks exposed east of the West fault and north of the Milagro Reverse fault. A regional tectonic event in the middle Permian exhumed Permian metamorphic and plutonic units and juxtaposed them structurally against coeval volcanic and sedimentary rocks prior to deposition of Triassic volcanic rocks and sediments (Tomlinson et al., 2012). This Permian deformation may explain the contrast in upper Carboniferous-lower Permian geology across the restored West fault, implying that at the latitude of the Calama basin, the West fault is a reactivated Permian structure.

Abundant alteration, veining with Fe and Cu sulfides, and Cu geochemical anomalies have been reported across the Ricardo property on both sides of the West fault. The most

significant of these are defined by a cluster of closely spaced drill holes on the east side of the West fault immediately north of the Sierra Limón Verde and by wide-spaced drill holes on the southwest side of the Ricardo property on the west side of the fault (Fig. 4). Alteration across both areas is reported as dominantly sericitic with local propylitic or argillic alteration. Despite the local presence of pyrite and/or chalcopyrite, no ore-grade mineralization is intersected.

Discussion

The temporal relationship between the Ministro Hales and Chuquicamata Intrusive Complexes

In porphyry samples with a large dispersion in CA-ID-TIMS zircon dates, the youngest zircon in each sample is commonly interpreted as the closest estimate of the emplacement age, and the extended zircon record reflects evolution of the underlying magma reservoir prior to emplacement (e.g., von Quadt et al., 2011; Large et al., 2020). At least two separate emplacement events are evident from the investigated porphyries of the Ministro Hales intrusive complex (i.e., in the Ministro Hales mine and north body; Fig. 2A). The first evidence of porphyry emplacement is a Mansa Mina porphyry at 36.263 ± 0.057 Ma. After a 610-k.y. apparent gap, a second generation of porphyry emplacement occurred from 35.653 ± 0.052 to 35.410 ± 0.080 Ma (including intrusions mapped as quartz porphyry and Mansa Mina porphyry). Despite the gap in emplacement ages, zircon crystallization was continuous for at least 1.5 Ma (Fig. 2A), suggesting a common long-lived underlying source reservoir for all Ministro Hales porphyries. The end of documented porphyry emplacement at Ministro Hales occurs about 0.5 Ma before the emplacement of porphyries in the RT Norte area of the Chuquicamata Intrusive Complex, but the zircon record between the Ministro Hales porphyries and the old generation of porphyries in the Chuquicamata Intrusive Complex is virtually uninterrupted (Fig. 2; see Virmond et al., 2024). This suggests that the Chuquicamata Intrusive Complex and the Ministro Hales Intrusive Complex are both expressions of the same long-lived magmatic system. This is further supported by the presence of marginally older zircon date outliers within the Chuquicamata Intrusive Complex (Virmond et al., 2024), which occur within the same range of dates found in the Ministro Hales domain (see KDE curves in Fig. 2B).

The Re-Os date obtained from Ministro Hales at around 35 Ma overlaps with the inception of molybdenite precipitation in RT Norte in the northernmost part of the Chuquicamata Intrusive Complex (Fig. 2). This further confirms that the multimillion-year magmatic-hydrothermal systems that formed Ministro Hales and the deposits associated with the Chuquicamata Intrusive Complex are, in fact, parts of the same system. Yet, the data also unambiguously show that both the two generations of porphyries and the mineralization at Ministro Hales are significantly older than those at Chuquicamata (by ca. 2 Ma; Fig. 2B), precluding a twinship of these two deposits in contrast to previous suggestions (e.g., Zentilli et al., 2021).

Reconstructing the displacement of the West fault

Beyond their geochronological similarities, the Ministro Hales and RT Norte deposits also share important geologic charac-

teristics. Both deposits feature background potassic alteration with Cu-Mo mineralization overprinted by structurally controlled phyllic to advanced argillic alteration zones associated with high-sulfidation sulfide assemblages (see also App. Figs. A2, A3). Although a more detailed geologic description of RT Norte is pending, the advanced argillic alteration of the deposit is less intensely developed than at Ministro Hales and hosts a significantly lesser proportion of the Cu mineralization, potentially indicating a somewhat deeper erosion level, consistent with the down-to-the-west sense for the vertical component of displacement across the West fault (Tomlinson and Blanco, 1997).

The distance of the RT Norte deposit away from the West fault (>1.2 km) indicates that Ministro Hales and RT Norte are two distinct porphyry centers and not parts of the same deposit offset by the West fault. However, given the spatio-temporal trend in the Chuquicamata Intrusive Complex, the close similarity of the Ministro Hales and RT Norte zircon and molybdenite record (Fig. 2), and the broadly comparable geology of these two deposits, we suggest that they were adjacent deposits in the Eocene, prior to their offset by post-ore sinistral movement of the West fault. Therefore, the Ministro Hales Intrusive Complex likely represents the northernmost and oldest part of the Chuquicamata Intrusive Complex.

The distance between the analyzed samples from RT Norte and Ministro Hales is ~20 km, which is significantly less than the proposed 35 km of movement on the West fault (Fig. 5; Dilles et al., 1997; Tomlinson and Blanco, 1997). The 35-km offset is justified by numerous, consistently matching plutonic, volcanic, and sedimentary units, guide beds, and stratigraphic and structural contacts on both sides of the West fault as far as 130 km north of RT Norte (Dilles et al., 1997; Tomlinson and Blanco, 1997). Yet, several observations suggest the fault offset may significantly decrease southward: (1) Our interpretation of the pre-Oligocene geology below the Calama basin independently suggests an ~20-km sinistral displacement of the Eocene Icanche Formation across the West and Limón Verde faults (Fig. 4), and (2) geochronological data (LA-ICP-MS U-Pb on zircon) from the Arkansas prospect (located on the east side of the fault ~35 km north of Ministro Hales; Fig. 5) returned ages of 43 to 41 Ma (Tomlinson et al., 2018), significantly older than those at Ministro Hales.

Part of the reduced offset of the West fault south of Calama can be explained by the opening of the Oligocene to Miocene Calama basin, elongating the eastern block in the north-south direction (Fig. 5; Tomlinson et al., 2018). Some of this north-south extension was likely centered over the area between Chuquicamata and Sierra Limón Verde and thus does not influence the offset between RT Norte and Ministro Hales. However, extension also marginally affected the northern part of the basin between El Abra and Chuquicamata, which would reduce offset along the West fault between RT Norte and Ministro Hales (Tomlinson et al., 2018). Postmineralization north-south contraction in the west block north of Ministro Hales would also reduce offset from RT Norte, but there is no evidence for such contraction. Between the Arkansas prospect and the El Abra mine (Fig. 5), a set of parallel, closely spaced, NW-SE-trending, sinistral strike-slip faults links into the West fault, and subsidiary north-south sinistral faults lie parallel to the West fault (Dilles et al., 1997; Tomlinson et

al., 2001). These faults likely dissipate some sinistral displacement on the West fault, such that the offset south of El Abra is less. If a similar array of NE-SW-trending faults extending from north of Ministro Hales to the south and southeast existed below cover, this would also contribute to reducing offset between the Chuquicamata Intrusive Complex and Ministro Hales and imply that Ministro Hales is located within a fault sliver (Fig. 5), as previously suggested by Zentilli et al. (2021). Our restoration of Ministro Hales and RT Norte deposits places the missing half of Chuquicamata ~20 km south of the active Chuquicamata mine, below the southern urbanized part of Calama City, in the northern part of the Ricardo project property (Fig. 4).

Available search space for the missing half of the Chuquicamata orebody

To date, exploration efforts in the southern part of Calama City have been limited for obvious reasons. One water well (CHUCA-7B) has been drilled in the vicinity of the proposed location of the orebody on the west side of the West fault (Fig. 4). This hole reached 211-m depth and penetrated only 1 m into the pre-Oligocene basement consisting of compact breccia with irregular limonite-filled fractures. Two holes drilled ~2 km north and south of CHUCA-7B (RRD-10 and R-3, respectively) intersected basement (over 202 and 83 m, respectively) with locally intense propylitic alteration and Cu values around 100 ppm. It is noteworthy that such alteration and weak Cu anomalies, although typical of the distal parts of porphyry copper deposits, are also widespread within the basement of the Ricardo project on both sides of the West fault. This makes their assignment to a causative porphyry center uncertain. Overall, the limited geologic information from below the southern part of Calama City is insufficient to rule out the existence of a Cu orebody, and the area remains underexplored.

Keeping in mind that alteration mineralogy has not been reported consistently over the exploration lifetime of the Ricardo property, it is also noteworthy that alteration halos of deposits associated with the Chuquicamata Intrusive Complex (i.e., Chuquicamata, Radomiro Tomic, Ministro Hales) are elongated along the West fault and have relatively small east-west footprints (e.g., Ossandón et al., 2001; Rivera et al., 2012). Additionally, the alteration and ore zoning in the Chuquicamata deposit suggest the West fault did not propagate through the middle of the deposit but rather its western margin (Ossandón et al., 2001), meaning the missing piece will likely have a smaller east-west dimension than that of Chuquicamata (i.e., less than 500 m). Thus, closely spaced (<500 m), inclined drill holes would be most suitable for locating any mineralization alongside the fault. In the north-south orientation, the West fault truncates the mineralization over a length of 3 km within the Chuquicamata open pit. This is likely the maximum north-south extension of the offset part of the deposit, which would fit well in between drill holes RRD-10 and R-3. Further, the likelihood that the missing half of the Chuquicamata orebody has a leached cap underlain by an enriched chalcocite blanket akin to that at the Chuquicamata deposit (Ossandón et al., 2001; Pinget, 2016) highlights the need for any drilling to penetrate the basement for a few hundred meters, significantly more than has been

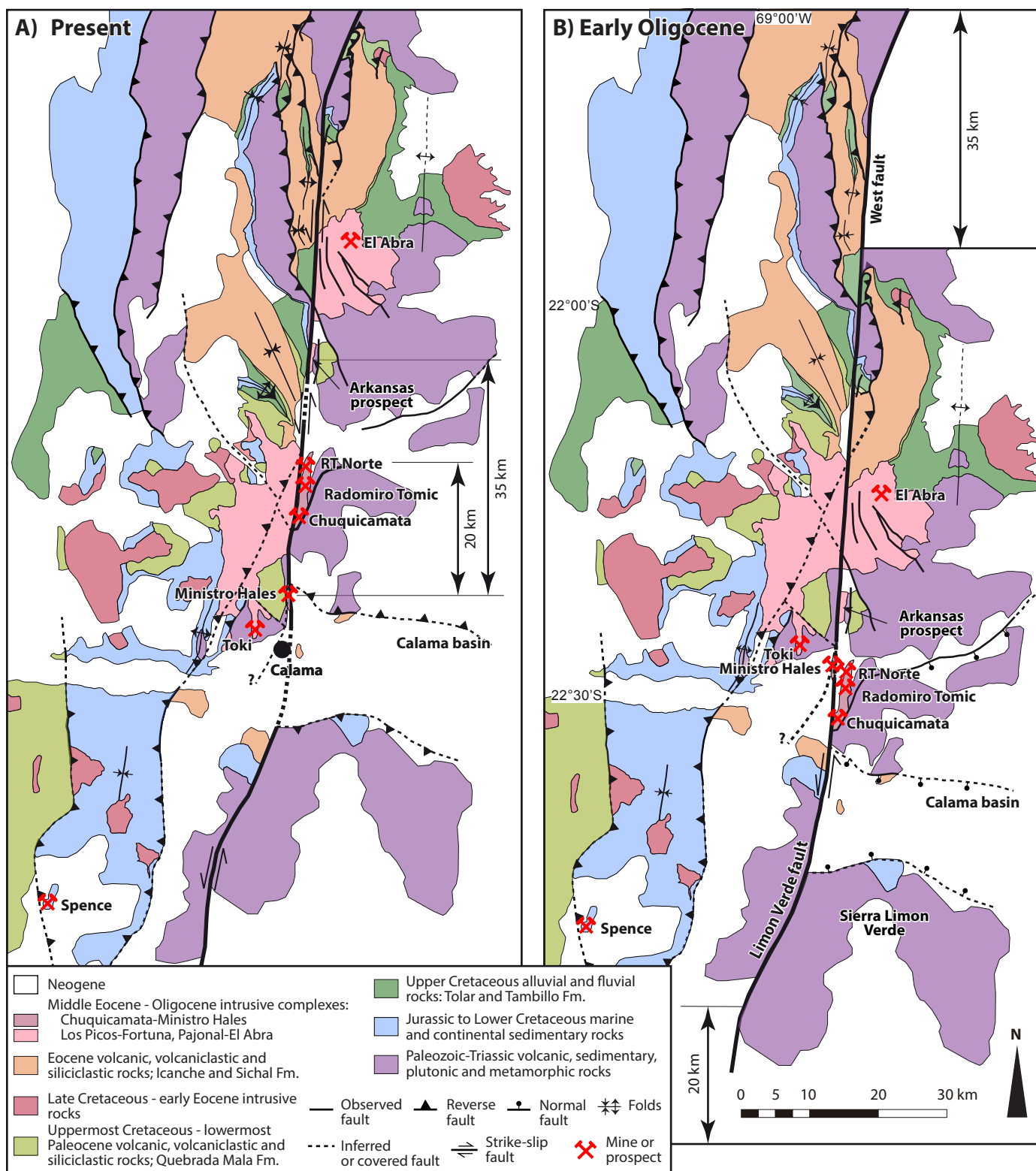


Fig. 5. Reconstruction of the displacement along the West fault. (A) Current. (B) Reconstruction at the time of magmatic-hydrothermal activity in the Chuquicamata Intrusive Complex. Modified after Tomlinson et al. (2019).

commonly done to date in the Ricardo property (only 16 of the 59 holes that intersected the basement units drilled over 200 m into it).

Conclusions

The timing of magmatic and hydrothermal activity associated with the Ministro Hales Intrusive Complex is constrained through high-precision zircon and molybdenite geochronology. The data reveal that the Ministro Hales porphyry system is among the earliest of the ore-forming magmatic-hydrothermal systems associated with the multimillion-year Chuquicamata Intrusive Complex, suggesting its location prior to post-ore sinistral displacement of the West fault was most likely in the vicinity of RT Norte. The corollary of this finding is that the missing western part of the Chuquicamata deposit, which could potentially host significant copper resources, may be located ~20 km to the south of the Chuquicamata mine, below the thick Oligocene to Quaternary sedimentary cover of the Calama basin. A similar offset of the orebody is independently supported by the interpreted pre-Oligocene geology that floors the Calama basin (Fig. 4). This would potentially put the deposit at depth below the southern part of Calama City, posing serious socioeconomic questions about the feasibility of any underground mining operation there.

Finally, we show that high-precision geochronology of magmatic and hydrothermal minerals can be an essential tool in reconstructing fault displacements. Robust dating of the mineralization proved to be key for reconstructing the greater system and should be useful for reconstructing other structurally dismembered ore deposits and potentially increasing the chances for discovery of buried orebodies.

Acknowledgments

ALV, DS, and CCM acknowledge funding from ETH Zürich. AJT and NB acknowledge support from SERNAGEOMIN. We are deeply grateful to the members of CODELCO for providing access to drill core samples and for sharing valuable insights into the geology of the Chuquicamata district. We thank Marcel Guillong and Lorenzo Tavazzani for laboratory support at the Department of Earth and Planetary Sciences at ETH Zurich. We acknowledge the support of Chris Ottley and Geoff Nowell and the use of the Durham Geochemistry Centre within the Department of Earth Sciences at Durham University. We are grateful for the insightful review by Richard Sillitoe, which prompted us to conduct an in-depth compilation of exploration data over the Ricardo property, to Travis Murphy for additional reviews, and to Dave Cooke for editorial handling. We are indebted to Sergio Rivera, Carolina Núñez, Carlos Münchmeyer, Guillermo Müller, Juan Carlos Marquardt, John Proffett, Marcelo Garcia, Teresa Jordan, and Eugenia Fonseca, who kindly shared data and information on the history of the Ricardo project.

REFERENCES

Alvarez, O., 1993, MM, the other side of Chuquicamata?: Society for Mining, Metallurgy, and Exploration, Annual Meeting, Reno, 1993, Preprint 93-139, 3 p.

Amilibia, A., Sàbat, F., McClay, K.R., Muñoz, J.A., Roca, E., and Chong, G., 2008, The role of inherited tectono-sedimentary architecture in the development of the central Andean mountain belt: Insights from the Cordillera de Domeyko: *Journal of Structural Geology*, v. 30, p. 1520–1539.

Arndt, N.T., Fontboté, L., Hedenquist, J.W., Kesler, S.E., Thompson, J.F.H., and Wood, D.G., 2017, Future global mineral resources: Geochemical Perspectives, v. 6, p. 1–171, <https://doi.org/10.7185/geochempersp.6.1>.

Baker, R.C., and Guilbert, J.M., 1987, Regional structural control of porphyry copper deposits in northern Chile [abs.]: *Geological Society of America Abstract with Programs*, v. 19, p. 578.

Barra, F., Alcota, H., Rivera, S., Valencia, V., Munizaga, F., and Maksae, V., 2013, Timing and formation of porphyry Cu-Mo mineralization in the Chuquicamata district, northern Chile: New constraints from the Toki cluster: *Mineralium Deposita*, v. 48, p. 629–651.

Boric, R., Díaz, J., Becerra, H., and Zentilli, M., 2009, Geology of the Ministro Hales mine (MMH), Chuquicamata district, Chile [ext. abs.]: XII Congreso Geológico Chileno, Santiago, Chile, 2009, Actas, S11_055.

Camus, F., and Castelli, J.C., 2021, Historia, exploración y geología de los yacimientos metalíferos de Chile 900-2021: Santiago, Origo Ediciones, 1338 p.

Camus, F., and Dilles, J.H., 2001, A special issue devoted to porphyry copper deposits of northern Chile: *Economic Geology*, v. 96, p. 233–237.

Cobbold, P.R., Rossello, E.A., Roperch, P., Arriagada, C., Gómez, L.A., and Lima, C., 2007, Distribution, timing, and causes of Andean deformation across South America: *Geological Society, London, Special Publications*, v. 272, p. 321–343.

CODELCO, 2022, Comprehensive report 2022: CODELCO, Chile, 256 p., https://www.codelco.com/sites/site/docs/20230607/20230607094540/m22_eng_L.pdf.

Crane, D., and Kavalieris, I., 2012, Geologic overview of the Oyu Tolgoi porphyry Cu-Au-Mo deposits, Mongolia: Society of Economic Geologists, Special Publication 16, p. 187–213.

Darin, M.H., and Dorsey, R.J., 2013, Reconciling disparate estimates of total offset on the southern San Andreas fault: *Geology*, v. 41, p. 975–978.

Dilles, J.H., Tomlinson, A., Martin, M.W., and Blanco, N., 1997, The El Abra and Fortuna Complexes: A porphyry copper batholith sinistrally displaced by the Falla Oeste [ext. abs.]: VIII Congreso Geológico Chileno, Antofagasta, Chile, 1997, Actas, v. 3, p. 1883–1887.

Duhart, P., Muñoz, J., Quiroz, D., Mestre, A., and Varas, G., 2018, Carta Sierra Gorda, Región de Antofagasta: Servicio Nacional de Geología y Minería, Carta Geológica De Chile, escala 1:100,000, SERNAGEOMIN, Santiago, Chile, no. 198.

Freeport McMoran South America Ltda., 2019, Ricardo. Servicio Nacional de Geología y Minería, Sistema de Información Geológica de Exploración (SIGEX), https://portalgeo.sernageomin.cl/SIGEX_2019/0284_2807_Ricardo.

Fu, B., and Avata, Y., 2007, Displacement and timing of left-lateral faulting in the Kunlun fault zone, northern Tibet, inferred from geologic and geomorphic features: *Journal of Asian Earth Sciences*, v. 29, p. 253–265.

Green, D., and Greig, C.J., 2013, Technical report on Ricardo project, Calama area, El Loa Province, Region II Antofagasta, Chile: Report for Lowell Copper Ltd., 61 p., filed with the Canadian Securities Administration, <https://www.sedarplus.ca/csa-party/records/document.html?id=0f7e97fd8974a1e696ed9280024081fd244ee726def2703041ff93085a26e846>.

Jordan, T., Lameli, C.H., Kirk-Lawlor, N., and Godfrey, L., 2015, Architecture of the aquifers of the Calama basin, Loa catchment basin, northern Chile: *Geosphere*, v. 11, p. 1438–1474.

Large, S.J.E., Wotzlaw, J.-F., Guillong, M., von Quadt, A., and Heinrich, C.A., 2020, Resolving the timescales of magmatic and hydrothermal processes associated with porphyry deposit formation using zircon U-Pb petrochronology: *Geochronology*, v. 2, p. 209–230.

Lawley, C.J.M., and Selby, D., 2012, Re-Os geochronology of quartz-enclosed ultrafine molybdenite: Implications for ore geochronology: *Economic Geology*, v. 107, p. 1499–1505.

Lindsay, D.D., Zentilli, M., and Rivera, J.R.D.L., 1995, Evolution of an active ductile to brittle shear system controlling mineralization at the Chuquicamata porphyry copper deposit, northern Chile: *International Geology Review*, v. 37, p. 945–958.

Maksae, V., and Zentilli, M., 1999, Fission track thermochronology of the Domeyko Cordillera, northern Chile: implications for Andean tectonics and porphyry copper metallogenesis: *Exploration and Mining Geology*, v. 8, p. 65–89.

Mathur, R., Ruiz, J., and Munizaga, F., 2000, Relationship between copper tonnage of Chilean base-metal porphyry deposits and Os isotope ratios: *Geology*, v. 28, p. 555–558.

McInnes, B.I.A., Farley, K.A., Sillitoe, R.H., and Kohn, B.P., 1999, Application of apatite (U-Th)/He thermochronometry to the determination of the

- sense and amount of vertical fault displacement at the Chuquicamata porphyry copper deposit, Chile: *Economic Geology*, v. 94, p. 937–947.
- Mpodozis, C., and Cornejo, P., 2012, Cenozoic tectonics and porphyry copper systems of the Chilean Andes: *Society of Economic Geologists, Special Publication 16*, p. 329–360.
- Mpodozis, C., Marinovic, N., Smoje, I., and Cuitiño, L., 1993, Estudio geológico-estructural de la Cordillera de Domeyko entre Sierra Limón Verde y Sierra Mariposas, Región de Antofagasta : Servicio Nacional de Geología y Minería, Informe Registrado IR-93-04, 282 p., 3 maps, 1:100,000 scale.
- Nickerson, P.A., and Seedorff, E., 2016, Dismembered porphyry systems near Wickenburg, Arizona: District-scale reconstruction with an arc-scale context: *Economic Geology*, v. 111, p. 447–466.
- Ossandón, G., Fréaut, R., Gustafson, L.B., Lindsay, D.D., and Zentilli, M., 2001, Geology of the Chuquicamata mine: A progress report: *Economic Geology*, v. 96, p. 249–270.
- Phillips, R.J., Parrish, R.R., and Searle, M.P., 2004, Age constraints on ductile deformation and long-term slip rates along the Karakoram fault zone, Ladakh: *Earth and Planetary Science Letters*, v. 226, p. 305–319.
- Pinget, M.-C., 2016, Supergene enrichment and exotic mineralization at Chuquicamata, Chile: PhD thesis, Geneva, Switzerland, University of Geneva, 182 p.
- Pizarro, J., 2011, Exploración con sondajes Proyecto Pulucktur, Distrito Codelco Norte: Memoria de título, Universidad de Chile: 74 p., <https://repositorio.uchile.cl/handle/2250/104116>.
- Richards, J.P., 2003, Tectono-magmatic precursors for porphyry Cu-(Mo-Au) deposit formation: *Economic Geology*, v. 98, p. 1515–1533.
- Rivera, S.L., Alcota, H., Proffett, J., Díaz, J., Leiva, G., and Vergara, M., 2012, Update of the geologic setting and porphyry Cu-Mo deposits of the Chuquicamata district, northern Chile: *Society of Economic Geology, Special Publication 16*, p. 19–54.
- Rojas-Arancibia, R., 2023, Propuesta de modelo y evolución estructural del yacimiento Radomiro Tomic, distrito Chuquicamata [ext. abs.]: XVI Congreso Geológico Chileno, Santiago, Chile, 2023, *Actas*, p. 801–804.
- Sepp, M.D., and Dilles, J.H., 2018, Structural evolution, vein orientation, and paragenesis of the Botija porphyry Cu-Mo-(Au) deposit, Panama: *Economic Geology*, v. 113, p. 857–890.
- Sillitoe, R.H., 1973, The tops and bottoms of porphyry copper deposits: *Economic Geology*, v. 68, no. 6, p. 799–815.
- Sillitoe, R.H., 1988, Epochs of intrusion-related copper mineralization in the Andes: *Journal of South American Earth Sciences*, v. 1, p. 89–108.
- Sillitoe, R.H., and Perelló, J., 2005, Andean copper province: Tectonomagmatic settings, deposit types, metallogeny, exploration, and discovery: *Economic Geology 100th Anniversary Volume*, p. 845–890.
- Sillitoe, H.R., Marquardt, J.C., Ramírez, F., Becerra, H., and Gómez, M., 1996, Geology of the concealed MM porphyry copper deposit, Chuquicamata district, northern Chile: *Society of Economic Geologists, Special Publication 5*, p. 59–69.
- Stavast, W.J.A., Butler, R.F., Seedorff, E., Barton, M.D., and Ferguson, C.A., 2008, Tertiary tilting and dismemberment of the Laramide arc and related hydrothermal systems, Sierrita Mountains, Arizona: *Economic Geology*, v. 103, no. 3, p. 629–636.
- Tomlinson, A., and Blanco, N., 1997, Structural evolution and displacement history of the West fault system, Precordillera, Chile: Part 2, Postmineral history [ext. abs.]: VIII Congreso Geológico Chileno, Santiago, Chile, 1997, *Actas*, v. 2, p. 1873–1878, v. 3, p. 1878–1882.
- Tomlinson, A., Blanco, N., García, M., et al., 2012, Permian exhumation of metamorphic complexes in the Calama area: Evidence for flat-slab subduction in northern Chile during the San Rafael tectonic phase and origin of the Central Andean gravity high [ext. abs.]: XIII Congreso Geológico Chileno, Antofagasta, Chile, 2012, *Actas*, v. 2, p. 212–214.
- Tomlinson, A.J., Dilles, J.H., and Maksaev, V., 2001, Application of apatite (U-Th)/He thermochronometry to the determination of the sense and amount of vertical fault displacement at the Chuquicamata porphyry copper deposit, Chile—a discussion: *Economic Geology*, v. 96, p. 1307–1309.
- Tomlinson, A.J., Blanco, N., Dilles, J.H., Maksaev, V., and Ladino, M., 2018, Carta Calama, Región de Antofagasta: Carta Geológica De Chile, escala 1:100,000, SERNAGEOMIN, Santiago, Chile, no. 199.
- Tomlinson, A.J., Blanco, N., and Dilles, J.H., 2019, Regional structural controls on localizing the Eocene-Oligocene Chuquicamata-El Abra-Toki porphyry copper cluster, northern Chile [ext. abs.]: *Society of Economic Geologists, SEG 2019, Santiago, Chile, 2019, Abstract 2435*.
- Virmond, A.L., Wotzlav, J.-F., Rojas-Arancibia, R., Selby, D., and Chelle-Michou, C., 2024, Multi-million-year magmatic and hydrothermal activity is key to the formation of supergiant to behemothian porphyry copper deposits: *Contributions to Mineralogy and Petrology*, v. 179, article 88, <https://doi.org/10.1007/s00410-024-02167-4>.
- von Quadt, A., Erni, M., Martinek, K., Moll, M., Peytcheva, I., and Heinrich, C.A., 2011, Zircon crystallization and the lifetimes of ore-forming magmatic-hydrothermal systems: *Geology*, v. 39, p. 731–734.
- Wang, S., Wang, C., Phillips, R.J., Murphy, M.A., Fang, X., and Yue, Y., 2012, Displacement along the Karakoram fault, NW Himalaya, estimated from LA-ICP-MS U-Pb dating of offset geologic markers: *Earth and Planetary Science Letters*, v. 337–338, p. 156–163.
- Zentilli, M., Boric, R., Heaman, L.M., Mathur, R., and Hanley, J.J., 2015, New developments on the geology of MMH: Is it the “missing half” of Chuquicamata? [ext. abs.]: XIV Congreso Geológico Chileno, La Serena, Chile, 2015, *Actas*, v. 2, p. 351–354.
- Zentilli, M., Maksaev, V., Boric, R., and Wilson, J., 2018, Spatial coincidence and similar geochemistry of Late Triassic and Eocene-Oligocene magmatism in the Andes of northern Chile: Evidence from the MMH porphyry type Cu-Mo deposit, Chuquicamata district: *International Journal of Earth Sciences*, v. 107, p. 1097–1126.
- Zentilli, M., Graves, M.C., Mathur, R., Hanley, J.J., Heaman, L.M., and Boric, R., 2021, Locating the “missing half” of the giant Chuquicamata porphyry copper deposit, Chile: *Society of Economic Geologists, Special Publication 24*, p. 69–85.

Adrianna Virmond is a geologist specializing in mineral resource systems and geochemistry. She obtained a BSc in geology and a MSc in geochemistry and petrology from the University of São Paulo, Brazil. During her PhD studies at ETH Zurich, Switzerland, her research focused on the processes controlling the formation and evolution of ore deposits, with an emphasis on trace element and isotope geochemistry of copper porphyry deposits from Chile.

